

Name: _____ Date: _____

Collaborators: _____ Lab Section: _____

POSTLAB EXERCISES (10 points)*Submit the postlab to the TA at the end of the lab.***Conversion of Mechanical Energy into Heat (10 points)****Question 3****2 points**

Fill in your data from the mechanical equivalence of heat experiment in Data Table 1. Convert measured resistance R to temperature T using Table 1 and/or eq. (7).

Data table 1: *Data for mechanical equivalence of heat.*

Time [s]	0	30	60	90	120	150	180	210	240
R [k Ω]									
N [revs]									
Temp [$^{\circ}\text{C}$]									
Time [s]	270	300	330	360	390	420	450	480	510
R [k Ω]									
N [revs]									
Temp [$^{\circ}\text{C}$]									
Time [s]	540	570	600	630	660	690	720	750	780
R [k Ω]									
N [revs]									
Temp [$^{\circ}\text{C}$]									

Next, plot temperature versus time from Data Table 1 in Graph 1. Draw two best-fit straight lines: one for the interval $t = 0$ to $t = t_{\text{stop}}$, and another for $t = t_{\text{stop}}$ to $t = 2t_{\text{stop}}$.

In Graph 1, mark the initial temperature T_{initial} , the peak temperature T_{peak} , and the final temperature T_{final} , as shown in Fig. 3. These three temperatures must be read from the two best-fit straight lines, not from the data points directly. The initial temperature T_{initial} is the y -intercept of the first line; the peak temperature T_{peak} is at the intersection of the two lines; and the final temperature T_{final} is the value of the second line at $t = 2t_{\text{stop}}$.

As usual, include the title and axis labels (with units) in your plot to receive full credit.

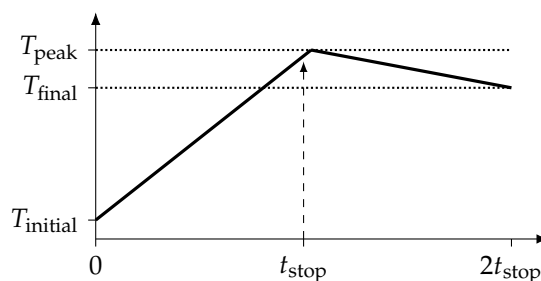
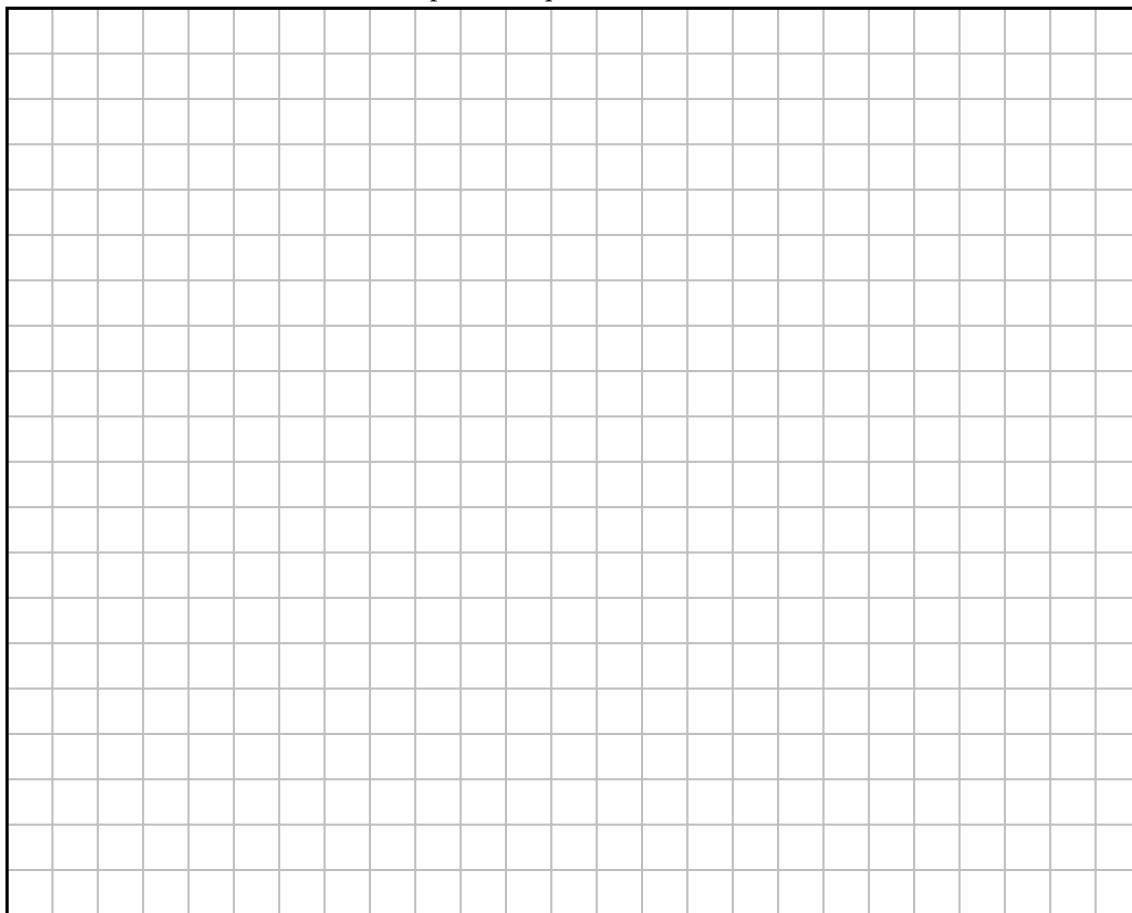


Figure 3: Expected time dependence of the temperature of the aluminum cylinder in this experiment.

Graph 1: Temperature vs. time.



Question 4**1 point**

The peak temperature in your graph may not coincide exactly with the temperature at t_{stop} . Why might the temperature reading continue to rise for a short time after you stop cranking?

Question 5**2 point**

(a) ($\frac{1}{2}$ point) Using eq. (3), calculate the work done, W , to lift the mass while cranking.

Total number of revolutions: $N =$ _____

Mass you lifted off the ground: $M =$ _____ kg

Mass of the aluminum cylinder: $m =$ _____ kg

Radius of the aluminum cylinder: $R =$ _____ m

(b) ($\frac{1}{2}$ point) Compute the change in temperature $\Delta T = T_{\text{peak}} - T_{\text{initial}}$.

(c) ($\frac{1}{2}$ point) Using eqs. (4) and (5), calculate the heat Q added to the system by raising its temperature by ΔT :

(d) ($\frac{1}{2}$ point) Using eq. (6), estimate Joule's constant. Don't forget units.

Question 6**1 point**

After you stop cranking, the aluminum cylinder loses heat to the environment because it is hotter than its surroundings. Calculate the temperature drop $\Delta T_{\text{lost}} = T_{\text{peak}} - T_{\text{final}}$.

Question 7**2 points**

Heat was also being lost to the environment while you were cranking (though the cylinder was cooler then than at T_{peak}). If we tried to correct for this by using $\Delta T + \Delta T_{\text{lost}}$ as the total temperature rise due to cranking, why would this likely overestimate the actual heat lost during cranking?

Question 8**2 points**

Discuss one major source of error in this experiment besides ambient cooling, and explain how it may have affected the measured Joule's constant. If that error were removed, would the measured Joule's constant increase or decrease? Explain.